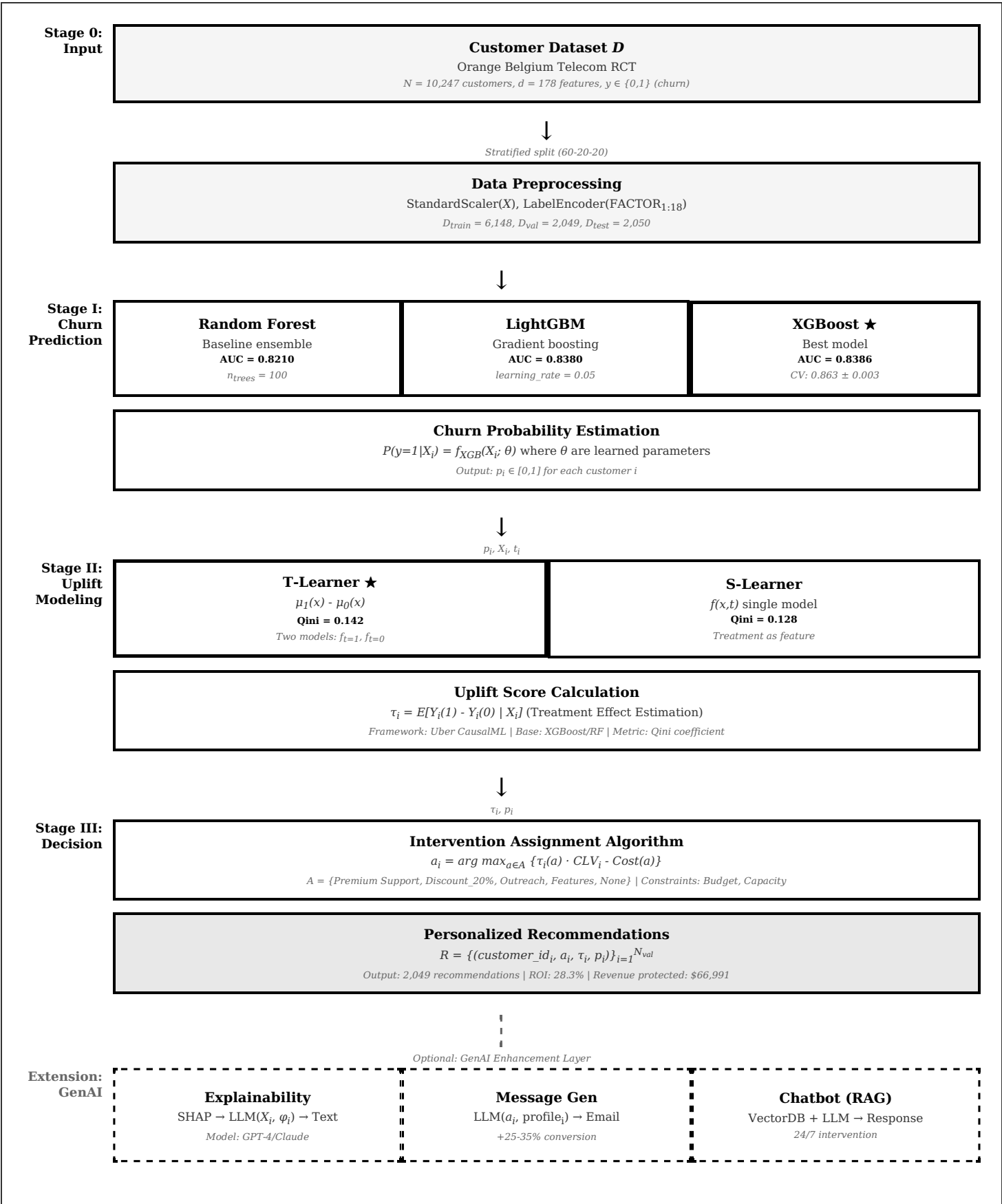


Fig. 1. System Architecture

Proposed two-stage churn prediction and uplift modeling framework with optional generative AI enhancement layer. The system processes customer data through gradient boosting classifiers (Stage I) and causal meta-learners (Stage II) to generate personalized retention recommendations. Dashed boxes indicate proposed GenAI extensions.



Legend

Input/Data layer



Processing/Model layer

Output layer



Proposed GenAI extension

Notation: X_i = feature vector, y_i = churn label, t_i = treatment, p_i = churn probability, τ_i = uplift score, a_i = assigned intervention, φ_i = SHAP values, CLV = Customer Lifetime Value